Quiz: GFS

Test your knowledge of the Google File System by completing this quiz.

Quiz ?

Statement

In GFS, files are divided into ------- chunks.

Select Multiple Options

False
- [ ] Variable size

Correct
- [x] Fixed size

False
- [ ] Both fixed and variable size

False
- [ ] None of the above

Check Answer

Correct Answer ✓

Quiz ?

Statement

To detect data corruption, GFS compares the contents of chunks on multiple ChunkServers.

Select Multiple Options

False

- [ ] True

Correct

- [x] False

Check Answer

Correct Answer

Quiz ?

Statement

File naming (or renaming) operation is atomic.

Select Multiple Options

Correct

- [x] True

False

- [ ] False

Check Answer

Correct Answer

Quiz ?

Statement

To collect ChunkServer's state, GFS master sends regular ------ messages to each ChunkServer.

Select Multiple Options

False ⊗

- [ ] KeepAlive

False ⊗

- [ ] Status

False ⊗

- [ ] StayAlive

Correct ✓

- [x] HeartBeat

Check Answer

Correct Answer ✓

Quiz ?

Statement

Does GFS use time-bound leases to reduce the network traffic?

Select Multiple Options

Correct

- [x] True

False

- [ ] False

Check Answer

Correct Answer

Quiz ?

Statement

Which two strategies are used to keep GFS highly available?

Select Multiple Options

False

- [ ] Fast Recovery and Checksumming

False

- [ ] Checksumming and Operation Log

False

- [ ] Checksumming and Master Replication

Correct

- [x] Fast Recovery and Chunk Replication

Check Answer

Correct Answer ✓

Quiz ?

Statement

Random writes into files in GFS are

Select Multiple Options

False ⊗

☐ Frequent

False ⊗

☐ Very Frequent

Correct ✓

☑ Less Frequent

False ⊗

☐ Rare

Check Answer

Correct Answer ✓

Quiz ?

Statement

Random writes into files in GFS are

Select Multiple Options

False

- [ ] Slow reads

Correct

- [x] Slow writes

False

- [ ] Slow appends

False

- [ ] None

Check Answer

Correct Answer ✓

Quiz ?

Statement

| **Distributing replicas onto different racks has following benefits (select all that apply)**:

Select Multiple Options

False

- [ ] Faster writes

Correct

- [x] Faster reads

Correct

- [x] Better reliability

False

- [ ] Data integrity

Check Answer

Correct Answer ✓

Quiz ?

Statement

What does a ChunkServer store? Select all that apply.

Select Multiple Options

Correct

- [x] Chunk data (one file per chunk)

False

- [ ] Chunk ACLs

Correct

- [x] Chunk metadata (version number, checksum)

False

- [ ] Location of chunk replicas

Check Answer

Correct Answer ✓